

LUCIDMSK CME PROGRAM

MODULE 01 — POST-TEST

Lumbar Spine MRI: Disc, Stenosis & Endplate Grading

Spine · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1

On sagittal T2, a lumbar disc shows inhomogeneous gray-black signal, lost NP/AF distinction, and moderately decreased height. Pfirrmann grade:

- A Grade II
- B Grade III
- C Grade IV
- D Grade V

2

A disc herniation where the dome is **WIDER** than the neck, remaining in continuity with the parent disc, is classified as:

- A Broad-based bulge
- B Protrusion
- C Extrusion
- D Sequestration

3

Axial T2 shows no CSF signal but nerve roots remain individually distinguishable. Schizas grade:

- A Grade A
- B Grade B
- C Grade C
- D Grade D

4

Which Modic type(s) are FDA-cleared targets for basivertebral nerve (BVN) ablation?

- A Type 1 only
- B Type 2 only
- C Types 1 and 2
- D Types 1, 2, and 3

5

Sagittal T1 shows completely obliterated perineural fat with nerve root deformation at L4-5 right foramen. Lee grade:

- A Grade 0
- B Grade 1
- C Grade 2
- D Grade 3

6

L4 anterolisthesis on L5 measures 60% anterior slip. Meyerding grade:

- A Grade II
- B Grade III
- C Grade IV
- D Grade V (spondyloptosis)

7

A subarticular (paracentral) disc herniation at L4-5 most commonly compresses which nerve root?

- A L3
- B L4
- C L5
- D S1

8

Modic Type 1 is best characterized by which MRI signal pattern?

- A T1 hypointense, T2 hypointense
- B T1 hyperintense, T2 suppressed on STIR
- C T1 hypointense, T2/STIR hyperintense
- D T1 hyperintense, T2 hyperintense

9

A disc fragment has lost all continuity with the parent disc and migrated caudally. Best term:

- | | |
|---|---------------------------|
| A | Protrusion with migration |
| B | Extrusion |
| C | Sequestration |
| D | Foraminal herniation |

10 Key distinguishing feature of discitis vs. Modic Type 1 on MRI:

- | | |
|---|---|
| A | Presence of T2 endplate signal alone |
| B | Disc height loss + T2 disc signal + paraspinal/epidural soft tissue |
| C | Subchondral sclerosis on T1 |
| D | Enhancement of endplates post-contrast |

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	C	C	C	B	C	C	C	B

Q1

On sagittal T2, a lumbar disc shows inhomogeneous gray-black signal, lost NP/AF distinction, and moderately decreased height. Pfirrmann grade:

Answer: C

A Grade II

B Grade III

✓ Grade IV

D Grade V

EXPLANATION Grade IV = inhomogeneous gray-black signal, lost NP/AF distinction, moderately decreased height. Grade V = completely collapsed disc space. Grade III has unclear (not lost) NP/AF.

Q2

A disc herniation where the dome is WIDER than the neck, remaining in continuity with the parent disc, is classified as:

Answer: C

A Broad-based bulge

B Protrusion

✓ Extrusion

D Sequestration

EXPLANATION Extrusion = dome wider than base (neck), maintained continuity. Protrusion = base wider than dome. Sequestration = lost continuity. Bulge = >180° symmetric extension, not a true herniation.

Q3

Axial T2 shows no CSF signal but nerve roots remain individually distinguishable. Schizas grade:

Answer: C

A Grade A

B Grade B

✓ Grade C

D Grade D

EXPLANATION Schizas C = no CSF but roots still identifiable as separate structures. Grade D = no CSF AND roots indistinguishable (solid block of tissue). Grade B = inhomogeneous but CSF still present.

Q4

Which Modic type(s) are FDA-cleared targets for basivertebral nerve (BVN) ablation?

Answer: C

A Type 1 only

B Type 2 only

✓ Types 1 and 2

D Types 1, 2, and 3

EXPLANATION Types 1 and 2 are the FDA-cleared indications for Intracept BVN ablation. Type 3 (sclerotic) is not a target. Type 1 has the strongest published pain response evidence.

Q5

Sagittal T1 shows completely obliterated perineural fat with nerve root deformation at L4-5 right foramen. Lee grade:

Answer: C

A Grade 0

B Grade 1

✓ **Grade 2**

D Grade 3

EXPLANATION Lee Grade 2 = completely obliterated fat + nerve root deformation. Grade 1 = partially obliterated without deformation. Grade 3 = nerve root entirely invisible within the foramen.

Q6

L4 anterolisthesis on L5 measures 60% anterior slip. Meyerding grade:

Answer: B

A Grade II

✓ **Grade III**

C Grade IV

D Grade V (spondyloptosis)

EXPLANATION Grade III = 51–75% slip. Grade II = 26–50%. Grade IV = 76–100%. Grade V = >100% (spondyloptosis, complete displacement). Grades III+ usually require instrumented fusion.

Q7

A subarticular (paracentral) disc herniation at L4-5 most commonly compresses which nerve root?

Answer: C

A L3

B L4

✓ **L5**

D S1

EXPLANATION Subarticular herniations compress the TRAVERSING root, one level below the disc level. L4-5 subarticular → L5 root. A FORAMINAL herniation at L4-5 compresses the EXITING L4 root.

Q8

Modic Type 1 is best characterized by which MRI signal pattern?

Answer: C

A T1 hypointense, T2 hypointense

B T1 hyperintense, T2 suppressed on STIR

✓ **T1 hypointense, T2/STIR hyperintense**

D T1 hyperintense, T2 hyperintense

EXPLANATION Type 1 = T1↓ hypointense, T2/STIR↑ hyperintense — active inflammatory vascular fibrous tissue. Type 2 = T1↑, STIR suppressed (fatty marrow). Type 3 = both hypointense (sclerosis).

Q9 A disc fragment has lost all continuity with the parent disc and migrated caudally. Best term:

Answer: C

A Protrusion with migration

B Extrusion

✓ **Sequestration**

D Foraminal herniation

EXPLANATION Sequestration = free fragment with NO continuity to parent disc. Extrusion retains continuity. Migration describes caudal travel. Always report direction and estimated distance.

Q10 Key distinguishing feature of discitis vs. Modic Type 1 on MRI:

Answer: B

A Presence of T2 endplate signal alone

✓ **Disc height loss + T2 disc signal + paraspinal/epidural soft tissue**

C Subchondral sclerosis on T1

D Enhancement of endplates post-contrast

EXPLANATION Discitis shows disc HEIGHT LOSS, T2 hyperintensity of the DISC ITSELF, and paraspinal/epidural extension. Modic 1 preserves disc height and lacks soft tissue spread. Both can show endplate enhancement.